

ArchiFlow AI — Database Architecture (Documented)

Document Control

- **System Name:** ArchiFlow AI
 - **Document Title:** Database Architecture (Logical + Physical)
 - **Version:** 1.0
 - **Date:** 2026-02-09
 - **Owner:** Project Team
 - **Scope:** Database design for user/workspace/project management, uploads, diagram generation, multi-LLM connections, versioning, documents (SRS/Journey/Blueprint), jobs, and audit.
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1. Objectives

This database architecture is designed to: 1. Store and manage **projects and diagrams** (flowcharts, system architectures, charts). 2. Support **uploads** (images/PDFs/other inputs) and their **extractions**. 3. Enable **multi-LLM usage** using user-connected accounts (ChatGPT/OpenAI, Microsoft Copilot, Google Gemini). 4. Provide **version control** for diagrams and documents. 5. Track **jobs** and **steps** for traceability, cost visibility, and troubleshooting. 6. Provide **audit trails** suitable for enterprise stakeholders.

2. Technology Stack Recommendation

2.1 Primary Database

- **PostgreSQL** (primary system of record)
- Structured entities: users, workspaces, projects, permissions, diagram metadata, versions, job tracking, audit.

2.2 File Storage (Non-Relational)

- **Object Storage (S3-compatible)**
- Stores large files: uploaded images/PDFs, generated outputs (PNG/SVG/PDF), exports.
- Database stores only **pointers** (bucket + key) and metadata.

2.3 Semantic Search (Optional, Phase 2)

- **pgvector in PostgreSQL** initially
 - For semantic search over prompts, extractions, diagram specs, documents.
 - Can later be swapped to a dedicated vector database if needed.
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3. Data Model Overview (Logical)

3.1 Tenancy Model

- Primary segregation is by **Workspace**.
- Most tables include `workspace_id` or are reachable via a `project_id` that is under a workspace.

3.2 Major Domains

1. **Identity & Access:** users, workspaces, membership, roles.
 2. **Project & Collaboration:** projects, members.
 3. **Assets & Extraction:** uploaded files, OCR/text/table extraction results.
 4. **Diagram Management:** diagrams, versions, inputs, rendering outputs.
 5. **AI Orchestration:** providers, connections, jobs, steps, prompts.
 6. **Documents:** SRS, user journey, blueprint, tech documents; versioned.
 7. **Governance:** audit events, retention policies.
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4. Entity Relationship Summary (ERD in Words)

- **Workspace 1—N Projects**
 - **Workspace 1—N Members (Users)**
 - **Project 1—N Assets**
 - **Asset 1—N Extractions**
 - **Project 1—N Diagrams**
 - **Diagram 1—N Diagram Versions**
 - **Diagram Version 1—N Inputs**
 - **Diagram Version 1—1 Rendered Output Asset (optional)**
 - **Workspace 1—N LLM Connections**
 - **Workspace/Project 1—N AI Jobs**
 - **AI Job 1—N Job Steps (optional)**
 - **Project 1—N Documents**
 - **Document 1—N Document Versions**
 - **Workspace/Project 1—N Audit Events**
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5. Table-by-Table Specification

Type Conventions - `uuid` used for primary keys. - `timestampz` for timestamps. - `jsonb` for flexible metadata. - Use **soft delete** (`deleted_at`) where appropriate.

5.1 Identity & Access

5.1.1 users

Stores user profile and authentication details. - **PK:** `id` - **Columns** - `id uuid` - `email text unique not null` - `name text` - `password_hash text` (nullable if OAuth-only) - `auth_provider text` (e.g., local/google/microsoft) - `status text` (active/suspended) - `created_at timestampz not null` - `last_login_at timestampz`

5.1.2 workspaces

Top-level tenant container. - **PK:** `id` - **Columns** - `id uuid` - `name text not null` - `owner_user_id uuid fk -> users.id` - `plan_tier text` (free/pro/enterprise) - `status text` (active/archived) - `created_at timestampz not null`

5.1.3 workspace_members

Membership + role. - **PK:** `id` - **Unique:** `(workspace_id, user_id)` - **Columns** - `id uuid` - `workspace_id uuid fk -> workspaces.id` - `user_id uuid fk -> users.id` - `role text` (Owner/Admin/Editor/Viewer) - `created_at timestampz not null`

5.1.4 api_keys (optional)

API keys for integrations. - **PK:** `id` - **Columns** - `id uuid` - `workspace_id uuid fk -> workspaces.id` - `label text` - `hashed_key text not null` - `scopes jsonb` (e.g., `{"projects:read":true}`) - `revoked_at timestampz` - `created_at timestampz not null`

5.2 Project & Collaboration

5.2.1 projects

- **PK:** `id`
- **Columns**
 - `id uuid`
 - `workspace_id uuid fk -> workspaces.id`
 - `name text not null`
 - `description text`
 - `status text` (active/archived)
 - `created_by uuid fk -> users.id`
 - `created_at timestampz not null`
 - `updated_at timestampz not null`

5.2.2 project_members (optional)

Project-level membership overrides. - **PK:** id - **Unique:** (project_id, user_id) - **Columns** - id uuid - project_id uuid fk -> projects.id - user_id uuid fk -> users.id - role text (Editor/Viewer) - created_at timestampz not null

5.3 Assets & Extraction

5.3.1 assets

Stores file metadata and object storage pointer. - **PK:** id - **Columns** - id uuid - workspace_id uuid fk -> workspaces.id - project_id uuid fk -> projects.id - type text (image/pdf/doc/json/csv) - filename text not null - mime_type text - size_bytes bigint - storage_bucket text not null - storage_key text not null - checksum_sha256 text - uploaded_by uuid fk -> users.id - uploaded_at timestampz not null

5.3.2 asset_extractions

Captures OCR/text/table extraction. - **PK:** id - **Columns** - id uuid - asset_id uuid fk -> assets.id - extraction_type text (ocr/text/table/metadata) - content_text text (or store pointer if huge) - status text (queued/running/success/failed) - error_message text - created_at timestampz not null

5.4 Diagram Management

5.4.1 diagrams

Top-level diagram record. - **PK:** id - **Columns** - id uuid - project_id uuid fk -> projects.id - title text not null - diagram_type text (flowchart/system_arch/chart/sequence/other) - source_mode text (prompt_only/from_upload/mixed) - current_version_id uuid fk -> diagram_versions.id (nullable during creation) - created_by uuid fk -> users.id - created_at timestampz not null - updated_at timestampz not null

5.4.2 diagram_versions

Version history + render output. - **PK:** id - **Unique:** (diagram_id, version_no) - **Columns** - id uuid - diagram_id uuid fk -> diagrams.id - version_no int not null - spec_format text (mermaid/plantuml/d2/json) - spec_text text not null - render_status text (queued/success/failed) - render_output_asset_id uuid fk -> assets.id (nullable) - created_by uuid fk -> users.id - created_at timestampz not null

5.4.3 diagram_inputs

What fed a specific version. - **PK:** id - **Columns** - id uuid - diagram_version_id uuid fk -> diagram_versions.id - input_type text (asset/prompt/context) - asset_id uuid fk -> assets.id (nullable) - prompt_text text (nullable) - order_no int not null

5.5 AI Orchestration (Multi-LLM)

5.5.1 llm_providers

Supported LLM vendors. - **PK:** id - **Columns** - id uuid - name text (OpenAI / Microsoft Copilot / Google Gemini) - provider_type text (openai/microsoft/google) - is_active boolean not null default true

5.5.2 llm_connections

Connected accounts for a workspace (personal/enterprise). - **PK:** id - **Columns** - id uuid - workspace_id uuid fk -> workspaces.id - provider_id uuid fk -> llm_providers.id - connection_name text - auth_type text (oauth/token) - token_encrypted text (or secret_ref if using vault) - token_expires_at timestampz - status text (active/expired/revoked) - created_by uuid fk -> users.id - created_at timestampz not null

5.5.3 ai_jobs

Async processing for extraction, generation, rendering, export. - **PK:** id - **Columns** - id uuid - workspace_id uuid fk -> workspaces.id - project_id uuid fk -> projects.id - job_type text (extract_asset/generate_diagram/refine_diagram/render_diagram/export) - status text (queued/running/success/failed/cancelled) - priority int not null default 5 - requested_by uuid fk -> users.id - started_at timestampz - finished_at timestampz - error_message text - correlation_id text - created_at timestampz not null

5.5.4 ai_job_steps (optional)

Step-level trace for debuggability and cost tracking. - **PK:** id - **Columns** - id uuid - job_id uuid fk -> ai_jobs.id - step_no int not null - step_type text (prompt_build/call_llm/parse/render) - status text (queued/running/success/failed) - provider_id uuid fk -> llm_providers.id - model_name text - tokens_in int - tokens_out int - cost_estimate numeric(12,6) - input_snapshot jsonb (must redact secrets) - output_snapshot jsonb (must redact sensitive) - started_at timestampz - finished_at timestampz

5.5.5 prompt_templates (recommended)

Reusable prompts for consistent outcomes. - **PK:** id - **Columns** - id uuid - workspace_id uuid fk -> workspaces.id (nullable for global) - name text not null -

purpose text - template_text text not null - variables_schema jsonb - is_active boolean not null default true - created_at timestampz not null

5.6 Documents (SRS / User Journey / Blueprint)

5.6.1 documents

Top-level records for generated or manually edited documents. - **PK:** id - **Columns:** id uuid - project_id uuid fk -> projects.id - doc_type text (srs/user_journey/blueprint/tech_spec/api_spec) - title text not null - current_version_id uuid fk -> document_versions.id (nullable) - created_by uuid fk -> users.id - created_at timestampz not null - updated_at timestampz not null

5.6.2 document_versions

Version history for all documents. - **PK:** id - **Unique:** (document_id, version_no) - **Columns:** id uuid - document_id uuid fk -> documents.id - version_no int not null - content_format text (markdown/html/json) - content_text text not null - generated_by_job_id uuid fk -> ai_jobs.id (nullable) - created_by uuid fk -> users.id - created_at timestampz not null

5.7 Governance & Audit

5.7.1 audit_events

Immutable activity log for compliance and troubleshooting. - **PK:** id - **Columns:** id uuid - workspace_id uuid fk -> workspaces.id - project_id uuid fk -> projects.id (nullable) - actor_user_id uuid fk -> users.id - event_type text (LOGIN/CREATE_PROJECT/UPLOAD_ASSET/GENERATE_DIAGRAM/DELETE_VERSION/EXPORT) - entity_type text - entity_id uuid - event_time timestampz not null - ip_address text - user_agent text - metadata jsonb

5.7.2 data_retention_policies (optional)

Retention rules per workspace. - **PK:** workspace_id - **Columns:** workspace_id uuid fk -> workspaces.id - asset_retention_days int - logs_retention_days int - version_retention_days int

6. Constraints & Rules (Non-Negotiables)

1. **Versioning** is mandatory for:
2. Diagrams (diagram_versions)
3. Documents (document_versions)

4. **Object storage pointers** must exist for every stored file (`assets.storage_bucket`, `assets.storage_key`).
 5. **Secrets** (LLM tokens) must never be stored in plain text.
 6. **Audit events** must record user actions impacting artifacts (uploads, generate, export, delete).
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7. Indexing Strategy (Minimum Required)

- `projects(workspace_id)`
 - `assets(project_id, uploaded_at DESC)`
 - `diagrams(project_id, updated_at DESC)`
 - `diagram_versions(diagram_id, version_no DESC)`
 - `ai_jobs(project_id, status, created_at DESC)`
 - `audit_events(workspace_id, event_time DESC)`
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8. MVP Scope vs Phase 2

8.1 MVP Tables

- `users`, `workspaces`, `workspace_members`
- `projects`
- `assets`, `asset_extractions`
- `diagrams`, `diagram_versions`, `diagram_inputs`
- `llm_providers`, `llm_connections`
- `ai_jobs`
- `documents`, `document_versions`
- `audit_events`

8.2 Phase 2 Enhancements

- `ai_job_steps` (deep observability)
 - `prompt_templates`
 - `data_retention_policies`
 - `pgvector` embeddings for semantic search
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9. Implementation Notes (Developer Guidance)

- Prefer **UUID PKs**.
 - Use `jsonb` only for flexible payloads; keep core fields relational.
 - Consider soft deletes for user-facing entities.
 - Enforce workspace boundaries at application and query layer.
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10. Appendix A — Naming & Status Enums (Recommended)

Roles

- Workspace roles: Owner, Admin, Editor, Viewer
- Project roles: Editor, Viewer

Job Status

- queued, running, success, failed, cancelled

Render Status

- queued, success, failed

LLM Connection Status

- active, expired, revoked